IPO Spillovers: Evidence from Private-Firm Networks

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November 10, 2026

Why do IPOs matter?

- Going public is a major event in a firm's life: it provides access to public equity markets and creates opportunities for early investors to realize their returns.
- Before an IPO, firms often spend years raising private capital from investors who finance several companies, creating networks of connected firms.
- When one company goes public, its IPO may reveal information about how public markets value the investors backing it, which can also matter for other firms connected through those investors.

Research question

Does one firm's IPO affect the subsequent IPO decisions of firms connected through common investors?

What This Paper Does

1. Construct a large dynamic private firm network based on common investors.
2. Study whether IPO decisions diffuse through these investor-linked firm networks.
3. Distinguish informational spillovers from correlated IPO activity generated by:
   - common investors, industry shocks, and hot IPO markets,
4. Develop a dynamic learning framework in which IPOs how markets value investors and generate spillovers across connected firms.

Related Literature

Literature	Relevant Papers
IPO timing and going-public decisions	(Zingales, 1995; Chemmanur and Fulghieri, 1999; Ritter and Welch, 2002; Ewens and Farre-Mensa, 2020)
Information asymmetry and IPO learning	(Hoffmann-Burchardi, 2001; Lowry and Schwert, 2002; Altı, 2005; Aghamolla and Guttman, 2021)
Investor reputation and certification	(Megginson and Weiss, 1991; Gompers, 1996; Nahata, 2008; Nanda et al., 2020)
VC investors, networks, and firm outcomes	(Hochberg et al., 2007; Bottazzi et al., 2008; Chemmanur et al., 2011; Eldar and Grennan, 2024)
Information transmission and peer effects	(Foucault and Fresard, 2014; Bramoullé et al., 2020; Bailey et al., 2022; Aghamolla and Thakor, 2022; Yan, 2024)

Main Findings

1. Private firms are more likely to go public after connected firms go public.
2. Spillovers are substantially stronger through common lead investors.
3. The effect attenuates with network distance.
4. Peer IPO exposure is associated with higher IPO valuations, share prices, and proceeds.

Data and sample

- Crunchbase, 2000–2023: firms, IPOs, funding rounds, and investor identities.
- The annual panel follows each firm only while it remains private.
- Post-IPO financing rounds do not create network links.

208,918 private firms

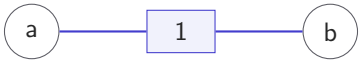
155,524 investors

5,422 IPOs among connected firms

The private firm network

Two firms are linked when they share at least one investor

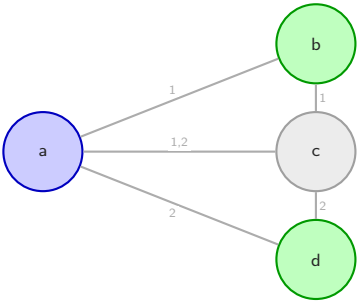
$$g_{ij,t} = 1 \iff K_{it} \cap K_{jt} \neq \varnothing.$$



common investor

- The network changes as firms raise new financing.
- Once observed, an investor connection remains part of the firm's financing history.

Peer IPO Exposure



Edge labels indicate the common investor(s).

$$IPOExposure_{i,t-1} = \frac{1}{|N_{i,t-1}|} \sum_{j \in N_{i,t-1}} IPO_{j,t-1}$$

- Firms are connected if they share at least one investor.
- Exposure is the share of direct peers that went public in the previous year.

$$IPOExposure_{\text{a},2021} = \frac{2}{3}$$

Empirical framework

$$IPO_{it} = \alpha + \delta IPOExposure_{i,t-1} + \gamma X_{i,t-1} + \rho_i + \lambda_{t \times industry \times country} + \varepsilon_{it}$$

- Outcome: whether focal firm i goes public in year t.
- Firm fixed effects absorb persistent differences across firms.
- Year $\times$ industry $\times$ country fixed effects absorb granular common shocks.
- Controls include lagged characteristics of the focal firm and its active peers.

Linear probability model. Standard errors clustered by firm.

Peer IPOs predict the focal firm's IPO

	No controls	Focal controls	All controls
Lagged IPO Exposure	0.017***	0.017***	**0.017*** **
	(0.006)	(0.006)	**(0.006)**
Firm FE	Yes	Yes	Yes
Year × industry × country FE	Yes	Yes	Yes
Observations	1,059,860	991,388	989,915

- A 10 p.p. increase in peer IPO exposure is associated with a **0.17 p.p. increase** in the annual IPO probability.

Spillovers are stronger through lead investors

	IPO
Lead-investor peer exposure	0.060***
	(0.022)
Non-lead peer exposure	0.013**
	(0.006)
p-value: equality	0.044

- Lead investors are more involved in governance and IPO preparation.
- Peer IPO exposure through these links has a substantially larger coefficient.

Spillovers within and across industries

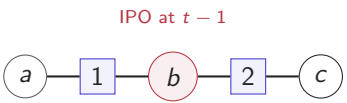
	IPO
Same-industry IPO Exposure	0.018***
	(0.004)
Other-industry IPO Exposure	0.011**
	(0.006)
p-value: equality	0.304

- Both coefficients are positive and are not statistically different.
- The pattern extends beyond conventional industry peers.

Prior second-order IPO exposure attenuates the direct-peer effect

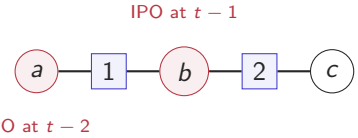
$$IPO_{i,t} = \alpha + \delta_{\text{noprior}} \sum_{j \in N_{i,t-1}} \widetilde{g}^{S=0}_{ij,t-1} IPO_{j,t-1} + \delta_{\text{prior}} \sum_{j \in N_{i,t-1}} \widetilde{g}^{S=1}_{ij,t-1} IPO_{j,t-1}$$
$$+ \boldsymbol{\gamma} \mathbf{X}_{i,t-1} + \beta_{\text{noprior}} \sum_{j \in N_{i,t-1}} \widetilde{g}^{S=0}_{ij,t-1} \mathbf{X}_{j,t-1} + \beta_{\text{prior}} \sum_{j \in N_{i,t-1}} \widetilde{g}^{S=1}_{ij,t-1} \mathbf{X}_{j,t-1} + \rho_i + \mu_{t \times r \times s} + \varepsilon_{i,t},$$

No prior second-order IPO



$$\widehat{\delta}_{\text{no prior}} = 0.023^{***}$$
$$(0.008)$$

Prior second-order IPO



$$\widehat{\delta}_{\text{prior}} = 0.005^{**}$$
$$(0.002)$$

- Difference: 0.019; p-value of equality: 0.021.

Peer exposure and outcomes at IPO

$$Y_{it} = \alpha_i + \lambda_{sct} + \beta\,\mathrm{PeerIPOShare}_{i,t-1} + \gamma' X_{i,t-1} + \delta' Z_{i,t-1} + \varepsilon_{it}.$$

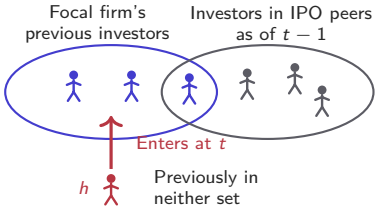
Y_{it}: log share price, valuation, IPO proceeds, or cumulative funding, observed for firms that go public.

Outcome	Prior peer IPO exposure
Log share price	0.247***
Log valuation	0.545***
Log IPO proceeds	0.445***
Log cumulative funding	0.287***

- Prior peer IPO exposure is positively associated with outcomes observed when the focal firm lists.

A new investor after peer IPO exposure

$$\text{NewNonIPOInvestor}_{it} = \alpha_i + \beta\,\text{PeerIPOShare}_{i,t-1} + \lambda_{sct} + \gamma' X_{i,t-1} + \delta' Z_{i,t-1} + \varepsilon_{it}.$$



Outcome = 1 if at least one such investor enters.

Number of investors shown is illustrative.

	New investor entry
Prior peer IPO exposure	0.0287*
	(0.0154)
Observations	989,915

- The estimate is less directly tied to liquidity released to shared investors by the peer IPO.

IPO peers: direct peers that went public in $t-1$. Linear probability model with lagged focal-firm and peer controls; firm and industry × country × year fixed effects. Standard errors clustered by firm. $^{*}p < 0.10$.

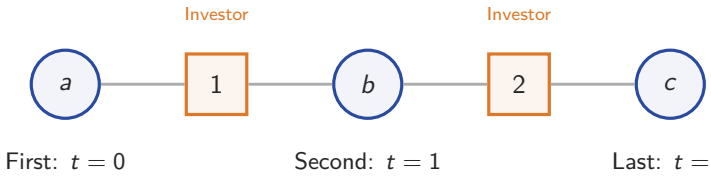
Identification evidence

Aggregate and local shocks	Firm FE and year $\times$ industry $\times$ country FE
Industry shocks	Separate exposure to same- and different-industry peers
Investor heterogeneity	Investor controls and exact cumulative leader-VC investor-set FE
Mechanical network correlation	Degree-preserving randomized networks
Any remaining correlated shocks	Policy (JOBS Act) as an IV

Recap

- Positive peer IPO spillovers.
- Stronger effects through common lead investors.
- Attenuation with network distance.
- Effects on IPO pricing and valuation outcomes.
- Higher probability of attracting new investors after peers IPOs.

Model Environment



- Firms $i \in \{a, b, c\}$; investors $k \in \{1, 2\}$.
- Fixed links; one predetermined IPO opportunity per firm.
- Three periods: $t = 0, 1, 2$. Firms move sequentially.
- Each decision becomes public before the next firm acts.

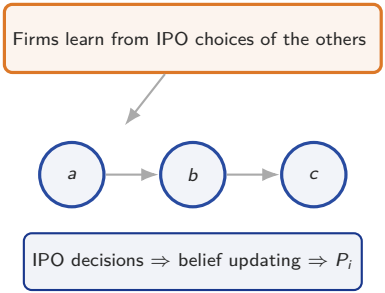
Notation in the appendix

Agents and Information

- **Firm value:** Expected firm quality plus the public market's valuation of the investors backing the firm.
- **Investor valuation:** For each investor $k \in \{1, 2\}$, the market's valuation ϵ_t^k evolves over time.
- **Initial information.** Before IPO decisions, the market knows how it values each investor, at each firm's IPO opportunity. Firms learn from signals about this value.
- **Firms.** Firm i chooses whether to IPO or not:

$$\mathbb{E}_i[P_i \mid \mathcal{I}_i] - \kappa_i \qquad \text{vs.} \qquad W_i.$$

- **Public market.** Given ϵ_i^k, the market sets price P_i under rational expectations with a pool of competitive and risk-neutral investors.
- **Each firm acts once: its date is implicit in P_i.**



Signal structure

Actions and Strategies

- **Given that each firm acts once: its date is implicit in action d_i.**

$$d_i \in \{0, 1\}, \qquad 1 = \text{IPO}, \quad 0 = \text{remain private}.$$

$$d_i = \sigma_i(\mathcal{I}_i) = 1 \quad \Longleftrightarrow \quad \mathbb{E}_i[P_i \mid \mathcal{I}_i] = \hat{P}_i \geq \kappa_i + W_i.$$

- P_i: public-market valuation; the firm forecasts it using its information $\mathcal{I}_i$.
- κ_i: IPO cost; W_i: value of remaining private.
- $\mathcal{I}_i$ contains initial private information and earlier observed decisions.
- σ_i specifies the action for every possible information set $\mathcal{I}_i$.

Information sets and strategies

Price Formation

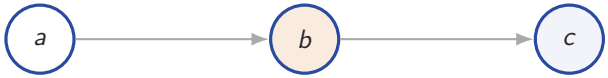
- The expected value by the market of a firm i is

$$V_i = \bar{\theta}_i + \sum_{k \in K_{it}} \alpha_i^k \epsilon_{it}^k, \qquad \bar{\theta}_i > 0.$$

- The public market defines the equilibrium price as

$$P_i = \mathbb{E}_M \left[V_i \mid \mathcal{I}_M \right].$$

Notation



$d_a = 0$: private $\qquad$ $d_b = 1$: IPO $\qquad$ c now decides

$$P_c\left(\underbrace{0}_{a\text{'s decision}}, \underbrace{1}_{b\text{'s decision}}\right); \qquad \big(\Delta_b(0,1) > 0\big).$$

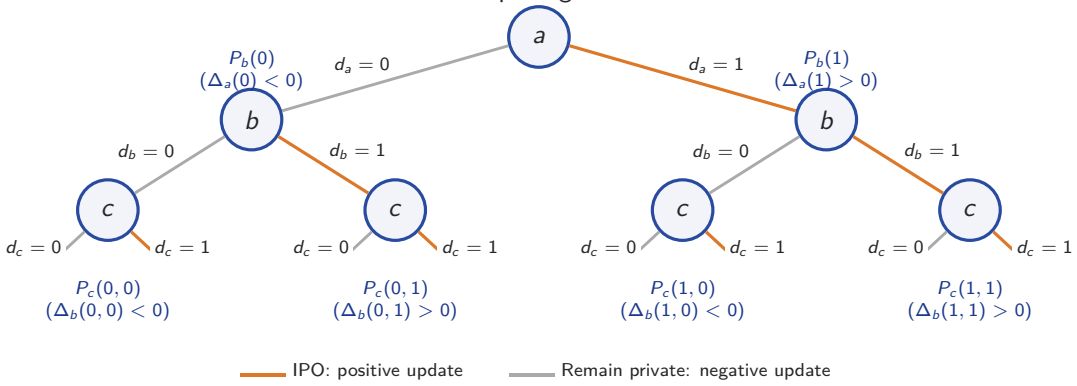
- $P_c(0,1)$ is firm c's price after these two decisions.
- $\Delta_b(0,1)$ is the price gain generated by firm b's decision, already included in that price.

Full tree

Illustration of the Sequential IPO Decisions

The figure represents only the sequential order of moves. Private signals are realized before firms act; information sets are omitted for readability.

Equilibrium concept: Perfect Bayesian Equilibrium (PBE), with competitive public-market pricing.



Arguments follow the decision order: $P_c(d_a, d_b)$. Δ parentheses show the price gain or loss generated by the peer.

Learning from a Direct Peer IPO



- Relative to non-IPO, b's IPO raises the market's assessment of investor 2.

$$P_c = \bar{\theta}_c + \alpha_c^2 \mu_2^2.$$

- μ_2^2: the market's belief about investor 2 when firm c decides.
- α_c^2: investor 2's equity share in firm c, which affects firm c's price.

$$d_b = 1 \quad \Longrightarrow \quad \mu_2^2 \uparrow \quad \Longrightarrow \quad P_c \uparrow .$$

Indexed valuation

Gains and losses

Indirect Peer Learning



Orange nodes indicate firms that completed an IPO.

Under the model's second-order information assumption, firm a's IPO:

1. raises firm b's expected IPO valuation;
2. reduces firm b's uncertainty.

There is no guarantee of normality when c decides on IPO, so we need to check if

$$\frac{Prob(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{Prob(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)}$$

is increasing in e. I therefore evaluate the attenuation condition numerically using deterministic integration.

Conclusions

1. IPO decisions propagate through firms networks formed by common investors.
2. Spillovers are stronger through common lead investors.
3. Peer IPO exposure affects both IPO timing and IPO outcomes.
4. The empirical patterns are consistent with firm learning about how market values investors.
5. This can explain the fact of fewer firms going IPO in the last decades Ewens and Farre-Mensa (2020) given that firms do not internalize the informational gains from IPO on peers.

Thank you

Questions and comments welcome.

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Backup: Presentation Notation and Full Indices

Each firm acts once. The main slides omit its decision date:

	Main slides	Fully indexed notation
First firm's action	d_a	d_{a0}
Second firm's action	d_b	d_{b1}
Last firm's action	d_c	d_{c2}
Last firm's price	P_c	P_{c2}
Private value	W_i	W_{it}
Firm information	$\mathcal{I}_f$	$\mathcal{I}_{it}$
Investor i's equity share in f	α_i^k	α_{it}^k

- Firm-specific i denotes the firm's IPO opportunity.
- Time remains explicit in investor ability and beliefs; equity shares are fixed.

Return to model

Backup: Investor Value and Signals

- **Public market value assigned to investor when evaluating firm i, for $k \in \{1, 2\}$:**

$$\epsilon_0^k \sim \mathcal{N}(0, \sigma^2_{\epsilon,i})$$

- **Firm signal, received only at $t = 0$:**

$$s_{i0}^k = \epsilon_0^k + \eta_{i0}^k, \qquad \eta_{i0}^k \sim \mathcal{N}(0, \sigma^2_\eta),$$

Return to agents

- ϵ_{it}^k are iid across investors k.
- $Cov(\epsilon_{it}^k, \epsilon_{jt+1}^k) > 0$

Backup: Information Sets and Market Beliefs

$$\mathcal{I}_{a0} = \{X_a, s^1_{a0}\},$$
$$\mathcal{I}_{b1} = \{X_b, s^1_{b0}, s^2_{b0}, d_{a0}\},$$
$$\mathcal{I}_{c2} = \{X_c, s^2_{c0}, d_{a0}, d_{b1}\}.$$
$$\mathcal{I}_{1M} = \{\mathcal{I}_{0M}, d_{a0}\}, \qquad \mathcal{I}_{2M} = \{\mathcal{I}_{1M}, d_{b1}\}.$$

- $\mathcal{I}_{0M}$ contains observed characteristics and its valuation about investors.
- Firms forecast prices by integrating out market-only signals.
- Strategies: $d_{it} = \sigma_f(\mathcal{I}_{it})$; monotone PBE combines optimal choices and Bayesian beliefs.

Return to actions

Backup: Firm Component and Prices

$$\bar{\theta}_i = \mathbb{E}_M[\theta_i \mid X_i].$$

Both firms and the market know the firm's prior fundamental value $\bar{\theta}_i$; neither directly observes true fundamental value θ_i.

$$\begin{aligned} P_a &= \bar{\theta}_a + \alpha_a^1 \varepsilon_a^1, \\ P_b &= \bar{\theta}_b + \alpha_b^1 \varepsilon_b^1 + \alpha_b^2 \varepsilon_b^2, \\ P_c &= \bar{\theta}_c + \alpha_c^2 \varepsilon_c^2. \end{aligned}$$

- α_i^k is investor k's equity share in firm i.
- Initial market signals are implicit in all history-dependent expressions.

Return to agents

Backup: What the Gains in the Tree Mean

The decision of ℓ changes the price of m:

$$\Delta_a(a_{a0}) = P_{b1}(d_{a0}) - \mathbb{E}_M[P_{b1} \mid \mathcal{I}_{0M}].$$

The decision of b changes the price of c:

$$\Delta_b(d_{a0}, d_{b1}) = P_{c2}(d_{a0}, d_{b1}) - \mathbb{E}_M[P_{c2} \mid \mathcal{I}_{1M}].$$

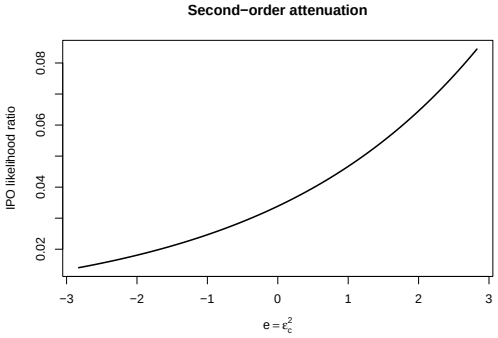
- Each gain is the change relative to the expected valuation before observing that peer's action.
- Under informative monotone learning, with both actions possible:

$$\Delta_a(0) < 0 < \Delta_a(1), \qquad \Delta_b(d_{a0}, 0) < 0 < \Delta_b(d_{a0}, 1).$$

- Negative Δ means a valuation loss from the news, not a negative total firm value.

Return to tree Return to direct learning

Backup: Conditional Attenuation



Relative selectivity of b's IPO decision

Numerical exercise

There is no closed-form solution, so I evaluate the sufficient condition for attenuation numerically.

The figure plots

$$\frac{\Pr(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 0)}{\Pr(d_b = 1 \mid \varepsilon_c^2 = e, s_c^2, d_a = 1)}.$$

The ratio is **increasing in** ε_c^2 over the evaluated support, consistent with the sufficient condition for **second-order attenuation**.

Benchmark: $\sigma_\varepsilon^2 = \sigma_\eta^2 = 1$, $\mathrm{Cov}(\varepsilon_a^1, \varepsilon_b^1) = \mathrm{Cov}(\varepsilon_b^2, \varepsilon_c^2) = 0.6$, $\alpha_a^1 = 1$, $\alpha_b^1 = \alpha_b^2 = 0.5$, $\bar{\theta}_a = \bar{\theta}_b = 1$, $W_a + \kappa_a = 2.98$, $W_b + \kappa_b = 2.4$, $s_c^2 = 0$.

Return to indirect learning

Interpreting the IPO Threshold

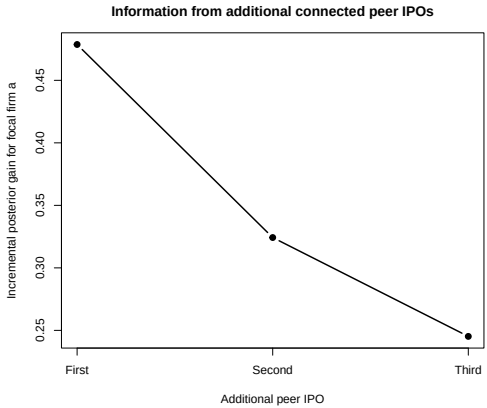
$$z_{b1}(d_{a0}) = \frac{W_b + \kappa_b - \mathbb{E}_c[\widehat{P}_b \mid s_c^2, d_a]}{\sqrt{\mathrm{Var}_c(\widehat{P}_b \mid s_c^2, d_a)}}. \tag{1}$$

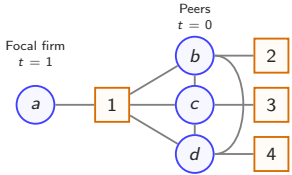
- z_{b1} measures the standardized distance to firm b's IPO threshold.
- The market forms the conditional mean and variance of the firm's expected IPO price.
- $z_{b1} > 0$: the threshold is above that mean; $z_{b1} < 0$: it is below.

Backup: Lead Investors and Several Direct Peers

- **Lead investors:** a larger equity share α_i^k amplifies the effect, provided the IPO is at least as informative about the common investor.
- **Several common investors and Direct overlap:** more common investors i increase by more the Δ_i when holding the total investor-base size, the peer's decision rule, and investor-specific updates fixed.
- **Several IPO peers that are dependent**: it might exist redundancy in the information that additional peers add, but still a positive spillover.

Network Structure and Informational Redundancy





Pure relevant cliques

- IPO peers are mutually connected exclusively through investors relevant to the focal firm.
- These IPOs may convey overlapping information.

$$\beta_{\text{non-pure}} = 0.110^{***}$$

$$\beta_{\text{pure}} \approx 0.028$$

Peer IPO spillovers are significantly weaker in pure relevant cliques.

Backup: Baseline Controls

Control	Definition
Transformed firm age	Age transformation used in the regression (`lnage2`).
Cumulative investments	Number of investment events, including repeated investments.
Cumulative unique investors	Number of distinct investors that have financed the firm.
Average investor score	Investors' historical IPO-exit share, averaged across the firm's investors.
Cumulative lead-investor share	Share of cumulative investment observations involving lead investors.
Investor-industry specialization	Share of each investor's other portfolio firms in the focal firm's industry, averaged across its investors.

- All controls enter the baseline lagged by one year.
- Peer controls: corresponding averages across active direct peers; peer age is measured in levels.